Bilal Aytekin

College Park, MD | baytekin@umd.edu | bilalaytekin.github.io

RESEARCH INTERESTS

Number theory, arithmetic geometry, and formalization of mathematics; in particular, motivic height zeta functions and weighted projective stacks.

EDUCATION

University of Maryland, College Park 2023 – present
Ph.D. in Mathematics College Park, MD

- Advisor: Dori Bejleri.

Boğaziçi University 2018 – 2023
B.Sc. in Mathematics and B.Sc. in Computer Engineering (double major) Istanbul, Turkey

- Thesis (Computer Engineering): *Entropy Hierarchy*. Advisor: Cem Say.

PUBLICATIONS AND PREPRINTS

1. **Higher-genus motivic height zeta functions for weighted projective stacks**.
   In preparation.
2. **MathAdv: What Theorem Provers Know, Reason, Formalize, and Generalize**.
   With J. Yuan, C. Martinez Lockhart, X. Liu, J. Wang, C. Deng, X. Han, V. Mastrantonis, D. Gudin, S. Zhu, A. Mohamed, J. Lang, Z. Song, and F. Huang. Submitted, 2026. arXiv:2608.25449.
3. **An Alexandrov Topology for Maximal Cohen–Macaulay Modules**.
   With M. Akdenizli, B. Çetin, and Ö. Esentepe. *Journal of Pure and Applied Algebra* 228 (2024), no. 1. arXiv:2210.03532.

TALKS

First Chern Classes Mar 2026
Student 6-Functor Formalism Seminar, University of Maryland

From Quadratic Functions to Modular Functions Dec 2025
Student Algebra and Number Theory Seminar, University of Maryland

Local and Global Root Numbers of Elliptic Curves Jun 2022
Summer School on Applied Arithmetic

Torsion Subgroups of Elliptic Curves Sep 2021
Directed Reading Program Turkey Symposium

TEACHING

University of Maryland 2023 – present
Department of Mathematics College Park, MD

Instructor

Introduction to Math Modeling and Probability	Fall 2025

Teaching Assistant

Abstract Algebra I (graduate)	Fall 2026
Introduction to Abstract Algebra	Fall 2026
Cryptography	Summer 2026
Linear Algebra for Scientists and Engineers	Spring 2026
Differential Equations	Spring 2025
Calculus II	Fall 2024
Calculus III	Spring 2024

Calculus II — Fall 2023

Boğaziçi University — 2022
Departments of Mathematics and Computer Engineering — Istanbul, Turkey
Undergraduate Assistant
Graduate Algebra II — Fall 2022
Grader
Introduction to Probability and Statistics for Computer Engineers — Fall 2022

CONFERENCES AND WORKSHOPS

Moduli of K3 Surfaces — Nov 2026
ICERM, Brown University, Providence, RI

Arithmetic and Algebraic Geometry Bootcamp 2026 — Aug 2026
Northwestern University, Evanston, IL

GAeL XXXIII: Géométrie Algébrique en Liberté — Jun – Jul 2026
Leiden, The Netherlands

JAVA 2026: Jeunes en Arithmétique et Variétés Algébriques — Jun 2026
Maison Clément, France

Algebraic Geometry Northeastern Series — May 2026
Rutgers University, New Brunswick, NJ

Arizona Winter School 2026: Computational Aspects of Arithmetic Geometry and Cryptography — Mar 2026
University of Arizona, Tucson, AZ

JAVA 2025: Jeunes en Arithmétique et Variétés Algébriques — Jul 2025
Maison Clément, France

Perspectives in Modern Mathematics: The Fifth Duke Mathematical Journal Conference — Apr 2025
Duke University, Durham, NC

Algebraic Geometry Northeastern Series — Nov 2024
Dartmouth College, Hanover, NH

Algebraic Geometry Northeastern Series — Oct 2023
University of Pennsylvania, Philadelphia, PA

AWARDS AND HONORS

Hauptman Fellowship — 2025 – 2026
Department of Mathematics, University of Maryland
Named for Nobel laureate Herbert A. Hauptman (Ph.D. 1955, Mathematics, University of Maryland).

Dean's Fellowship — 2023
The Graduate School, University of Maryland
Awarded through graduate programs to recruit and retain outstanding students.

COMPUTATIONAL SKILLS

Programming: Python, C++, Java
Formal verification: Lean 4
Computer algebra: SageMath, Magma
Languages: Turkish (native), English (fluent), French (intermediate)

Last updated November 2026.